



Learning with AI, Not from AI: A quasi-experimental comparison of AI-assisted learning with versus without an interactive AI-generated error-correction phase on knowledge application and metacognition

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ABSTRACT

Integrating generative AI into medical education raises concerns about students' uncritical acceptance of AI content, calling for strategies that foster critical evaluation. This quasi-experimental, pre-test–post-test study compared an interactive error-correction strategy in which students critiqued AI-generated answers to teacher-recommended and self-generated questions/statements with a control condition where students engaged in AI self-learning followed by teacher-led summarization and key point reinforcement on knowledge application and metacognitive skills in biomedical materials education. Medical students were assigned to a control group (n=90, AI self-learning + teacher summarization) or an intervention group (n=142, AI self-learning + peer correction of AI answers to teacher-recommended and student-generated questions/statements). After exclusions, 161 participants were analyzed (67 control, 94 intervention). The intervention group scored significantly higher on knowledge application (87.57 ± 6.50 vs. 78.22 ± 7.46 , $p < .001$) and metacognitive awareness (4.14 ± 0.58 vs. 3.78 ± 0.44 , $p < .001$). Secondary outcomes (engagement, critical evaluation) also favored the intervention. The AI-answer error-correction strategy improved short-term knowledge application and metacognitive awareness, offering a feasible approach to promote critical AI engagement in competency-based education.

1. Introduction

The integration of generative artificial intelligence (AI) into higher education has sparked considerable discourse regarding its potential to transform teaching and learning (Crompton et al. 2023; Zawacki-Richter et al. 2019). Technologies such as machine learning, natural language processing, and large language models (LLMs) are reshaping educational models (Bond et al. 2024; Kasneci et al. 2023), enhancing the quality and accessibility of instructional materials, and enabling personalized learning experiences (Bond et al. 2024; Moreno et al. 2023; Sapci et al. 2020). In disciplines such as medicine, nursing, and biomedical engineering, where complex conceptual knowledge must be applied to real-world problems, AI offers promise as a scalable educational resource (Akgun et al. 2022; Gordon et al. 2024). However, alongside these opportunities, significant concerns have emerged. Chief among them is the

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tendency for students to accept AI-generated content uncritically, a phenomenon often described as “AI authority bias” (Guo et al. 2023; Kim et al. 2023). When learners treat AI outputs as infallible, they risk bypassing the very cognitive processes that underpin deep learning, namely evaluation, verification, and reflection (Liu et al. 2025a; Stahl et al. 2024).

The challenge of fostering critical engagement with information is not new, but it acquires new urgency in the context of AI. Traditional pedagogical approaches, such as didactic lectures, typically position the instructor as the primary source of knowledge, with students assuming a receptive role (Broadbent et al. 2015). While effective for transmitting foundational content, such approaches may not explicitly cultivate the metacognitive skills required to navigate a landscape increasingly populated by AI-generated information (Kooli 2023; Ng et al. 2021). Metacognition, defined by Flavell as “the capacity to monitor, evaluate, and regulate one’s own cognitive processes” (Flavell 1979), is essential for self-regulated learning and has been shown to correlate with academic achievement across disciplines (Dahri et al. 2024; Zimmerman 2002). Yet, few instructional models deliberately leverage AI as a tool for metacognitive development.

One promising avenue is to re-conceptualize AI not as an authoritative answer engine but as a fallible learning partner (Kapur 2008; Sinha et al. 2021). By intentionally introducing AI-generated errors into learning activities, educators can create opportunities for students to engage in error detection, analysis, and correction, which are processes known to enhance conceptual understanding and cognitive engagement (McPhee et al. 2025; Metcalfe 2017). Error-based learning has a robust empirical foundation, with studies demonstrating that confronting and resolving errors promotes deeper knowledge integration and transfer (Kapur 2008; Sinha et al. 2021; Wong et al. 2022). Uniquely, the present intervention required students not only to detect errors in AI outputs but also to engage with both instructor-provided and self-generated questions/statements. This dual source of questions/statements encourages learners to take ownership of their learning by formulating their own inquiries could enhance conceptual clarity and metacognitive awareness. When applied to AI-generated content, this approach may simultaneously address two educational goals: improving domain-specific knowledge and cultivating the critical evaluation skills necessary for responsible AI use (Kooli 2023; Ng et al. 2021). Crucially, AI error-correction activities enhance metacognition by activating three core processes: monitoring, evaluation, and regulation, which are key components of self-regulated learning. Together, these processes directly strengthen the metacognitive capacities that underpin self-regulated learning.

2. Literature review

Despite the theoretical appeal of this approach, empirical evidence remains limited. Most studies on AI in education have focused on its efficacy as a direct tutor or its impact on student motivation and performance, with less attention to its role in developing metacognitive faculties (Dahri et al. 2024; Zawacki-Richter et al. 2019). Furthermore, controlled comparisons between AI-based interactive strategies and traditional instruction are scarce, particularly in the context of biomedical education (Gordon et al. 2024; Sapci et al. 2020).

2.1. Current research gaps and the paradigm shift in AI pedagogy

Recent reviews have mapped the evolving landscape of generative artificial intelligence (GenAI) in education with considerable clarity. For instance, Tu et al. identified a notable gap: while many investigations examine learners’ perceptions, relatively few analyse their cognitive and behavioural characteristics during actual GenAI use (Tu et al. 2025). Complementing this work, another 2025 synthesis integrated multidisciplinary perspectives and highlighted human-AI collaboration together with ethical considerations as central themes demanding further attention (Allison et al. 2025). Taken together, these contributions inform what has been termed the “Learning with AI, Not from AI” paradigm. This perspective reframes AI literacy beyond mere tool mastery toward shared agency and critical evaluation, suggesting that learners should interact with AI as a cognitive partner rather than an authoritative source. In this spirit, the present study attempts to address the identified gap by examining learners’ authentic problem-solving interactions with GenAI, thereby offering empirical insight into the theoretical discourse on student-AI collaboration.

2.2. Intervention context and design rationale

The present study evaluates an interactive AI-generated error-correction strategy in the context of biomedical materials education, specifically focusing on sterilization methods. This topic was selected for its high conceptual density and strong practical relevance, requiring learners to integrate principles of microbiology, materials science, and engineering design in a context where errors in AI summaries can be both subtle and consequential (Moreno et al. 2023). The intervention follows a structured sequence. First, students engage in inquiry-based learning by posing questions to AI. Second, teachers provide targeted supplementation to fill knowledge gaps and reinforce understanding. Third, students collaborate to identify, critique, and correct errors in AI-produced content. To isolate the effect of peer-driven error correction, the study compares this experimental condition against a control condition where students engage in AI self-learning followed by teacher-led summarization and key point reinforcement, without systematic error correction.

2.3. Study objectives and research questions

The primary objective of this study is to determine whether a peer-driven, AI-answer error-correction strategy, using both teacher-recommended and student-generated questions and statements, improves knowledge application and metacognitive awareness relative to a control condition of AI self-learning with teacher summarization. Building on the theoretical frameworks of error-based

learning (Kapur 2008; Metcalfe 2017) and self-regulated learning (Flavell 1979; Zimmerman 2002), we formulate the following research questions (RQ):

RQ1. Does the peer-driven AI-answer error-correction strategy result in significantly higher scores on a standardized knowledge application test compared with the control condition?

RQ2. Does the intervention lead to greater gains in overall metacognitive awareness, as measured by a validated inventory, and do these gains differ across the three metacognitive subscales (knowledge, monitoring, and regulation)?

RQ3. What are the secondary effects of the intervention on students' attitudes toward AI as a learning tool and their self-reported verification behaviors?

Consistent with these questions, we hypothesize (H) that: (a) the intervention group will outperform the control group on knowledge application (H1), drawing on evidence that error confrontation deepens conceptual integration; (b) the intervention group will report higher metacognitive awareness, with monitoring showing the largest effect size due to the immediate cognitive conflict inherent in comparing AI outputs against reliable references, followed by knowledge and regulation (H2), informed by metacognitive theory and recent AI literacy research; and (c) the intervention will cultivate more positive AI learning attitudes and more frequent verification behaviors relative to the control condition (H3), consistent with the "Learning with AI, Not from AI" paradigm.

3. Methods

3.1. Study design

This quasi-experimental, non-randomized educational evaluation was conducted from March to April 2026 at XXX Medical University, China. This study compared a control condition where students first learned with AI and then received teacher summarization and key point reinforcement with an experimental condition where students additionally engaged in peer correction of AI-generated answers to teacher-recommended and self-created questions/statements to enhance students' knowledge mastery and metacognitive competence in biomedical material sterilization. The study was conducted in accordance with the Declaration of Helsinki and the Institutional Research Ethics Committee of Jining medical university (JNMC-YX-2026-110). All participant information was kept strictly confidential, and electronic informed consent was obtained from all participants prior to data collection.

3.2. Participants inclusion and exclusion criteria

Participants were undergraduate students in their second semester of their second year, majoring in clinical medicine, who were enrolled in two parallel classes of the same course (same teachers but with slightly different class schedules). Students selected their class based on their schedule and personal preference. The control group (AI self-learning + teacher summarization) originally included 90 students, and the experimental group (AI self-learning + peer correction of AI answers) originally included 142 students. All participants had completed prerequisite courses in basic microbiology and had no prior formal training in sterilization technologies for biomedical materials. Inclusion criteria were voluntary participation and completion of all pre- and post-intervention assessments. Exclusion criteria included: (1) pre-test scores in the top or bottom 10% to minimize ceiling and floor effects; (2) absence from any intervention session or assessment point; (3) incomplete questionnaire or test responses.

3.3. Assessment time points

Standardized assessments were administered at fixed time points throughout the study. A knowledge pre-test on sterilization concepts (**Supplementary Table 1**) was administered to both groups three weeks before the intervention, and an identical knowledge post-test was administered two weeks after the intervention. The AI learning attitude and metacognitive attitude questionnaires were administered to both groups immediately after the intervention.

3.4. Intervention procedures

Both groups received a 80-minute classroom session covering identical core content, including sterilization concepts, spore resistance, thermal, chemical, radiation, and filtration sterilization methods, material-method compatibility, and clinical application schemes. The same instructor taught both groups to eliminate instructor-related bias.

Control group: AI Self-Learning plus Teacher Summarization and Key Point Reinforcement. The control group first completed a 50 minute AI-supported inquiry learning phase. Students worked in small groups (8-10 per group) using a structured task sheet with predefined AI prompts (**Supplementary Table 2**) and had the freedom to choose their preferred AI chatbots to explore core concepts, organize findings, and resolve confusion. They were also provided with PowerPoint slides as a reference. Following this, the instructor led a 30 minute summarization session, during which key concepts were highlighted, common misconceptions clarified, and the core content of sterilization methods summarized. The teacher also addressed any remaining student questions. No AI answer generation or error-correction activities were included in this condition.

Experimental Group: AI-Assisted Learning with Peer Correction of AI-Generated Answers. Phase 1: AI-supported inquiry learning

(50 minutes). This phase was identical to that of the control group. Students worked in small groups (8-10 per group) using the same structured task sheet and AI chatbots to explore core concepts. Phase 2: Peer correction of AI answers to teacher-recommended and self-generated questions/statements (30 minutes). In this phase, each student group (8-10 per group) first received a set of 3 questions/statements from the instructor, which covered key learning objectives of sterilization methods. Additionally, each group was asked to generate 2–3 of their own questions/statements related to the same topic. The groups then input these questions/statements (both teacher-recommended and self-generated) into their AI chatbot and recorded the AI-generated answers. Each group carefully reviewed the received answers, identified any factual errors, omissions, or inaccuracies, and corrected them using textbooks and reliable references. After the correction task, groups shared their findings and corrections with the class, discussing common AI mistakes and strategies for verification. This design required students to critically evaluate AI outputs in response to both externally provided and self-generated questions/statements, thereby deepening their understanding and enhancing metacognitive monitoring. Detailed step-by-step examples and actual screenshots of the AI chatbot interfaces are provided in **Supplementary Figure 1**.

3.5. Outcome measures

Sterilization Knowledge Acquisition: A 11-item standardized test was administered to both groups (**Supplementary Table 1**). The test consisted of ten single-choice questions (5 points each, total 50 points) and one comprehensive application question (50 points, for post-test only), with a total score of 50 and 100, respectively. The comprehensive question was scored using a validated rubric covering method selection, rationale, process parameters, precautions, and justification for excluding inappropriate methods. **Metacognitive Ability:** A 12-item scale adapted from the Metacognitive Awareness Inventory ([Schraw et al. 1994](#)) was administered to both groups after the intervention (**Table 1**). The scale measured three dimensions: metacognitive knowledge (4 items), metacognitive monitoring (4 items), and metacognitive regulation (4 items), using a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). **AI Learning Attitude:** A 10-item self-developed questionnaire using a 5-point Likert scale was administered to both groups after the intervention (**Table 2**). For the control group, who had no direct AI experience, the questionnaire assessed general attitudes toward AI rather than experience-based attitudes. The items assessed perceived reliability of AI, information verification behavior, critical awareness, learning dependence tendency, and willingness to continue using AI.

3.6. Statistical analyses

Data processing and statistical evaluation were conducted using GraphPad Prism and SPSS. Descriptive statistics characterized categorical variables as frequencies and percentages, while continuous variables were summarized using means ± standard deviations (SD). For questionnaires, reliability was assessed using Cronbach’s α, structural validity was examined via confirmatory factor analysis (CFA) for the adapted metacognitive scale and exploratory factor analysis (EFA) for the self-developed AI learning attitude scale. For comparative analyses, independent t-tests with Welch’s correction were used to compare the two groups, while analysis of covariance (ANCOVA) was applied to the post-test results. Effect sizes were calculated as Cohen’s *d* using the pooled standard deviation, with values of 0.2, 0.5, and 0.8 interpreted as small, medium, and large effects, respectively. Significance threshold was shown as **p* < .05, ***p* < .01. Scale reliability was assessed via Cronbach’s α coefficients.

4. Results

4.1. Demographic data

This study employed a quasi-experimental, non-randomized, pre-test–post-test design within two parallel classes. The intervention procedures are presented in **Fig. 1A**. Initially, the control class consisted of 90 students, while the experimental class comprised 142 students. Among these students, 6 from the control group and 25 from the experimental group declined to participate in the study. Consequently, pre-tests were administered only to the remaining 84 students in the control group and 117 students in the experimental

Table 1
Metacognitive awareness inventory (1 = strongly disagree, 5 = strongly agree).

Number	Question	Answer
1	Metacognitive knowledge	I can clearly tell how well I understand a concept while learning it.
2		I know which learning strategies work best for me.
3		I can judge whether a particular concept is important to my field of study.
4		I am aware of the situations in which I am likely to make mistakes.
5	Metacognitive monitoring	During the learning process, I continuously check whether I am keeping up with the content.
6		When I encounter difficulty in understanding, I pause and reconsider.
7		I can recognize when I have misunderstood a concept.
8	Metacognitive regulation	While reading or listening, I ask myself, “Am I really understanding this?”
9		When I realize that my understanding is off track, I actively adjust my learning strategies.
10		I adjust my learning pace according to the difficulty of the task.
11		When one approach proves ineffective, I try other methods.
12		After completing a learning task, I reflect on which strategies were effective and which were not.

Table 2

Questionnaire on attitudes toward using AI in learning (1 = strongly disagree, 5 = strongly agree).

Number	Question	Answer
1	I find AI information quite useful and use it as a starting point.	
2	I usually cross-check AI-generated answers against other sources or my own existing knowledge.	
3	When using AI, I consciously consider whether its responses are reasonable.	
4	AI tools help me learn more effectively (e.g., by enabling me to grasp concepts faster and identify gaps in my understanding).	
5	I think AI can be a good tool for preview or review, and even replace some teacher tasks.	
6	Even if AI provides a detailed solution, I will still reorganize and express it in my own words.	
7	Answers provided by AI often inspire me to raise new questions, which drive me to investigate topics more deeply.	
8	I believe that using AI as a long-term learning partner will broaden my intellectual horizons rather than limit them.	
9	I have become increasingly adept at articulating my actual needs to the AI in order to obtain more precise responses.	
10	When AI fails to provide a satisfactory answer, I know where to look in order to find a breakthrough direction.	

group. To minimize the potential ceiling and floor effects, participants whose pre-test scores fell within the top 10% and bottom 10% of each group's score distribution were subsequently excluded (Fig. 1B). After these exclusions, the final analytical sample comprised 67 students in the control group and 94 students in the experimental group (Fig. 1C). All of these participants completed the subsequent intervention and assessment procedures and were included in the final data analysis.

4.2. Pre-test and post-test knowledge scores

Prior to the intervention, no statistically significant difference was observed between the control and experimental groups on the knowledge pre-test, which consisted of ten single-choice questions (5 points each, total 50 points), with scores of control: 22.54 ± 6.93 and experimental: 22.87 ± 6.41 ($p > .05$), indicating baseline equivalence (Fig. 2A). Following the intervention, the post-test, comprising the same ten single-choice questions and an additional comprehensive application question (50 points), was administered with a total score of 100 points. After controlling for pre-test scores as a covariate, ANCOVA revealed a significant main effect of group on post-test knowledge scores, $F(1, 158) = 68.357$, $p < .001$, partial $\eta^2 = 0.302$. The adjusted post-test mean was significantly higher in the intervention group (87.61, 95% CI [86.19, 89.03]) than in the control group (78.18, 95% CI [76.42, 79.94]), with a mean difference of 9.43 (95% CI [7.17, 11.69]), with the Cohen's $d = 1.35$, $p < .001$ (Fig. 2B). These results suggest that the interactive

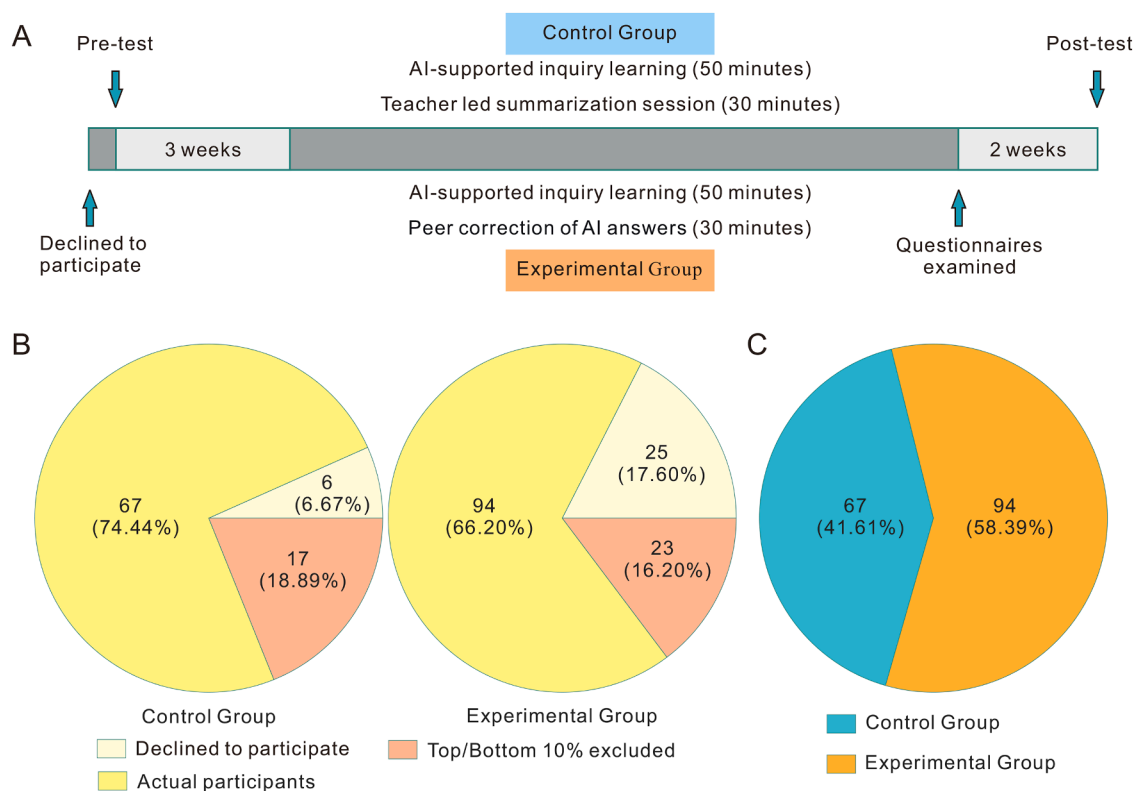


Fig. 1. Overall experimental arrangement and demographic information. (A) Flow diagram of the study. **(B)** Number/proportion of participants who voluntarily declined participation or were excluded. **(C)** Final numbers and percentages of participants included in the study.

AI-answer error-correction strategy was more effective than the control condition (AI self-learning plus teacher summarization without error correction) in improving students' knowledge application on sterilization methods for biomedical materials.

4.3. Usability analysis of survey questionnaires

Prior to formal analysis, the reliability and structural validity of both questionnaires were examined. For the adapted Metacognitive Ability Scale, descriptive statistics showed a score of Mean = 3.99 (SD = 0.55), 95% CI [3.903, 4.074], and Cronbach's $\alpha = 0.953$ (Table 3). Structural validity was assessed via confirmatory factor analysis (CFA), which supported the hypothesized three-factor model: $\chi^2/df = 1.820$, CFI = 0.968, RMSEA = 0.051, SRMR = 0.042, indicating acceptable convergent validity (Table 3). For the self-developed AI Learning Attitude Scale, the score was Mean = 3.95 (SD = 0.81), 95% CI [3.911, 3.991], with Cronbach's $\alpha = 0.891$. For structural validity, exploratory factor analysis (EFA) using principal component extraction with varimax rotation was performed. The Kaiser-Meyer-Olkin (KMO) measure was 0.879, and Bartlett's test of sphericity was significant ($\chi^2 = 795.5$, $df = 45$, $p < .001$) (Table 3). Collectively, these results support the adequate reliability and structural validity of both questionnaires, confirming their suitability for subsequent analyses.

4.4. Metacognitive ability results

Following the intervention, the experimental group demonstrated significantly higher metacognitive awareness scores than the control group (experimental: 4.14 ± 0.58 , control: 3.78 ± 0.44 , $t(158) = 4.385$, $p < .001$, Cohen's $d = 0.670$) (Table 4; Supplementary Table 4). Analysis of the three subscales revealed that the experimental group outperformed the control group across all dimensions: metacognitive knowledge (experimental: 4.08 ± 0.62 vs. control: 3.70 ± 0.47 , $t(158) = 4.357$, $p < .001$, Cohen's $d = 0.667$), metacognitive monitoring (experimental: 4.16 ± 0.59 vs. control: 3.74 ± 0.50 , $t(154) = 4.889$, $p < .001$, Cohen's $d = 0.760$), and metacognitive regulation (experimental: 4.17 ± 0.60 vs. control: 3.91 ± 0.49 , $t(155) = 3.060$, $p < .003$, Cohen's $d = 0.473$) (Table 4; Supplementary Table 4). These findings indicate that the interactive AI-answer error-correction strategy was more effective than the control condition in enhancing students' metacognitive capacities, particularly in monitoring and regulating their own learning processes.

4.5. Changes in AI learning attitude

The AI learning attitude questionnaire was administered to both the control group and the experimental group after the intervention. An independent-samples t-test indicated a statistically significant difference in overall attitude scores between the experimental and control groups, with the experimental group outperforming the control group (experimental: 4.15 ± 0.60 vs. control: 3.67 ± 0.37 , $t(155) = 6.146$, $p < .001$). The magnitude of this difference was large, as reflected by Cohen's $d = 0.910$ (Fig. 3;

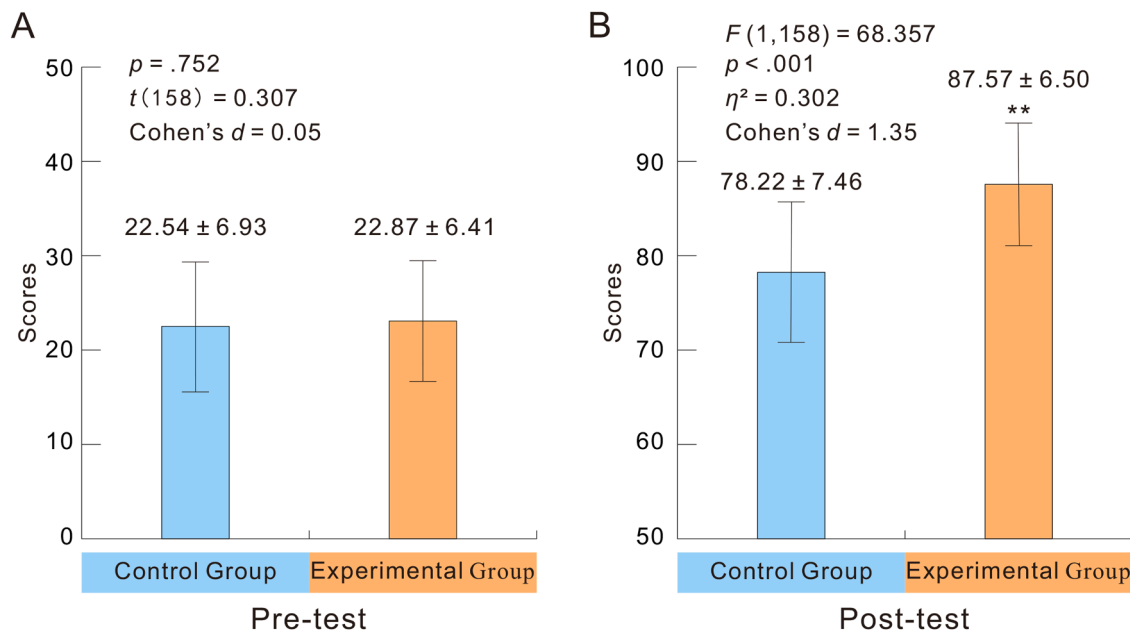


Fig. 2. Knowledge scores of pre-test and post-test. (A) Comparison of pre-test knowledge scores between the control and experimental groups ($p > .05$), demonstrating baseline equivalence. (B) Comparison of post-test knowledge scores between the control and experimental groups ($p < .001$), indicating a significant improvement in the experimental group.

Table 3

Reliability and validity of the metacognitive awareness inventory and AI learning attitude scale.

	Mean $\pm$ SD	95% CI	α	χ^2/df	CFI	RMSEA	SRMR
Metacognitive Awareness	3.99 $\pm$ 0.55	3.903, 4.074	0.953	1.820	0.968	0.051	0.042
AI attitude	3.95 $\pm$ 0.81	3.911, 3.991	0.891	0.879	795.5	45	< .001

Table 4

Group differences across all dimensions related to metacognitive awareness inventory.

Category	Control (n=67) Mean $\pm$ SD	Experimental (n=94) Mean $\pm$ SD	<i>t</i> value	<i>df</i>	<i>p</i> value	Cohen's <i>d</i>
Total (Q1 – Q12)	3.78 $\pm$ 0.44	4.14 $\pm$ 0.58	4.385	158	< .001	0.670
Knowledge (Q1 – Q4)	3.70 $\pm$ 0.47	4.08 $\pm$ 0.62	4.357	158	< .001	0.667
Monitoring (Q5 – Q8)	3.74 $\pm$ 0.50	4.16 $\pm$ 0.59	4.889	154	< .001	0.760
Regulation (Q9 – Q12)	3.91 $\pm$ 0.49	4.17 $\pm$ 0.60	3.060	155	.003	0.473

Supplementary Table 5). These results indicate that the interactive error correction strategy positively shifted students' attitudes toward AI, fostering more critical and self-regulated engagement.

5. Discussion

This study examined whether a peer-driven, AI-answer error-correction strategy could improve knowledge application and metacognitive awareness relative to a control condition where students engaged in AI self-learning followed by teacher summarization and key point reinforcement. The results indicate that students who critiqued and corrected AI-generated answers to teacher-recommended and self-generated questions/statements scored higher on a standardized knowledge application test and reported higher metacognitive awareness than their peers in the control group. As detailed in the Results section, the intervention yielded a moderate-to-large effect size on the knowledge application test (Cohen's $d = 0.667$) and consistent gains across all metacognitive subscales (Cohen's d ranging from 0.473 to 0.760), providing a comprehensive empirical basis for the effectiveness of the training program. These findings carry theoretical and practical implications, particularly as generative AI tools become increasingly prevalent in medical and engineering education.

From a theoretical perspective, the observed advantage of the peer correction of AI answers is consistent with established principles of constructivist and error-based learning (Banihashem et al. 2025). Cognitive theories of learning have long emphasized that confronting errors, rather than passively receiving correct answers, prompts deeper cognitive engagement. This process compels learners

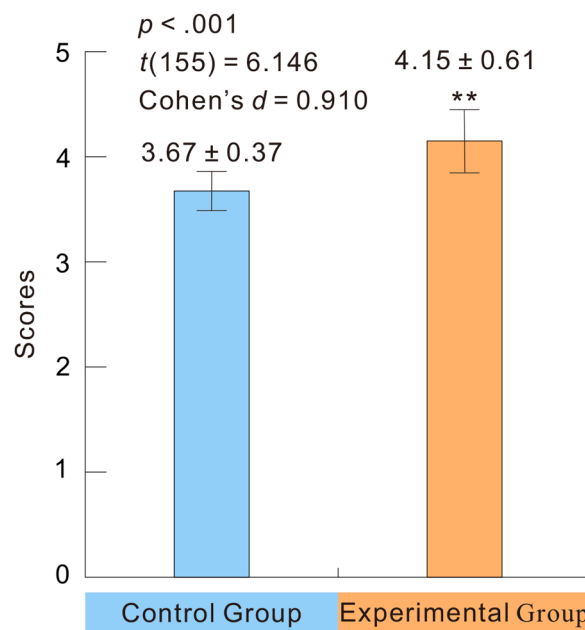


Fig. 3. Comparison of AI learning attitudes. Overall attitude scores between the control and experimental groups ($p < .001$), indicating a significantly more positive attitude in the experimental group.

to reexamine their prior knowledge, resolve contradictions, and construct more robust mental models (Kapur 2008; Sinha et al. 2021). In the present study, participants in the intervention group did not merely treat AI outputs as authoritative summaries; instead, they were required to scrutinize content, identify inaccuracies, and articulate justifications for corrections. This pattern of activity mirrors the productive failure paradigm, in which learners benefit from grappling with problematic material before receiving expert guidance (Metcalfe 2017). By embedding errors within AI generated content, the intervention leveraged the inherent fallibility of current large language models, thereby transforming a potential liability into a deliberate pedagogical asset (Liu et al. 2025b). Extending this line of reasoning, recent scholarship has called for a paradigmatic shift from transmissive pedagogy to relational learning in the AI era, emphasizing that educational principles, pedagogical theories, and technological frameworks should be explicitly articulated to bridge learning gaps (Nussbaum et al. 2025; Nussbaum et al. 2026). Our intervention embodies such a shift by repositioning AI as a corrigible partner rather than an authoritative answer engine, thereby fostering a relational dynamic in which students actively negotiate meaning with and through AI-generated content. Our finding that the intervention group reported significantly more positive AI learning attitudes (4.15 ± 0.61 vs. 3.67 ± 0.37 , $p < .001$, Cohen's $d = 0.910$) is consistent with this framework, suggesting that metacognitive engagement fostered by the error-correction task may have contributed to more favorable attitudes toward AI as a learning tool.

The higher metacognitive awareness scores observed in the intervention group relative to the control group further underscore the value of repositioning AI as a corrigible partner rather than an infallible source. Metacognition, defined as the capacity to monitor, evaluate, and regulate one's own cognitive processes, is widely recognized as a cornerstone of self-regulated learning and clinical reasoning (Flavell 1979; Zimmerman 2002). Teacher summarization, while helpful for reinforcing key concepts, does not explicitly activate the same level of error detection and regulatory processing as the peer-correction task (Broadbent et al. 2015). Importantly, the intervention produced differential effects across metacognitive subscales: monitoring showed the largest improvement (Cohen's $d = 0.760$), followed by knowledge (Cohen's $d = 0.667$) and regulation (Cohen's $d = 0.473$). The pronounced gain in monitoring is theoretically expected, as the error-correction task required continuous comparison of AI outputs against reliable references, directly engaging evaluative vigilance (McPhee et al. 2025). The moderate gain in metacognitive knowledge likely reflects the articulation and justification of corrections, which externalizes strategic heuristics (Schraw et al. 1994). The smallest yet significant gain in regulation aligns with theoretical accounts positioning regulation as a higher-order executive process that consolidates over repeated practice (Flavell 1979; Zimmerman 2002). As McPhee argue, fostering critical AI literacy requires a circular pedagogical framework across multiple cycles precisely because regulatory skills demand iterative reinforcement (McPhee et al. 2025). Liu et al. similarly found that AI-supported self-assessment remains underutilized even among high-exposure students (Liu et al. 2025b). Notably, these processes were rendered socially visible through group work, enabling learners to observe and internalize the critical evaluative strategies of their peers, a dimension often absent in individualized AI use (Joo et al. 2018; Simsek et al. 2025). Complementing these findings, recent empirical work has demonstrated that self-regulated learning skills, cognitive flexibility, and metacognitive awareness positively affect students' attitudes toward generative AI, with metacognitive awareness acting indirectly through the former two constructs (Yilmaz et al. 2025). Furthermore, cognitive flexibility, digital competencies, and self-regulation skills collectively reduce students' AI anxiety, suggesting that supportive learning environments should introduce AI tools as facilitative resources rather than sources of apprehension (Karaoglan Yilmaz et al. 2025). Our intervention, by embedding structured peer correction within a collaborative setting, directly cultivates these enabling skills and thereby reduces the potential for AI-related anxiety.

The findings also contribute to ongoing discussions surrounding concerns regarding "AI authority bias", that is, the tendency for students to accept AI-generated information uncritically (Guo et al. 2023; Kim et al. 2023; Martín-Moncunill et al. 2025). By intentionally having students critique AI answers to both teacher-recommended and self-created questions/statements, the intervention provided a structured, low-stakes environment in which to challenge AI authority (McPhee et al. 2025). This experience appears to have shifted students' epistemic stance toward generative AI, encouraging them to approach such tools with prudent skepticism and a heightened inclination to verify information. The secondary outcome measures support this interpretation: following the activity, students in the intervention group reported more frequent verification behaviors and a greater awareness of AI limitations compared with control group (Martín-Moncunill et al. 2025). In an era where large language models are increasingly used for self-directed learning, cultivating such calibrated trust is essential. Rather than discouraging AI use, the AI-answer error-correction approach equips students with the critical evaluative competencies necessary for responsible and effective AI utilization.

These results extend the literature on AI-enhanced education in several ways. Previous research has largely focused on AI's role as a direct tutor or its impact on motivation and academic performance (Gordon et al. 2024; Luo et al. 2025; Zawacki-Richter et al. 2019). This study shows that a peer-based AI-answer correction activity leads to better knowledge application and metacognitive outcomes than a control condition involving AI self-learning with teacher summarization (no error correction). Accordingly, the findings offer empirical justification for integrating such strategies into competency-based curricula. Therefore, this study provides empirical justification for integrating similar strategies into competency-based curricula. Moreover, the intervention design, which used teacher-recommended and student-generated questions/statements, offers a scalable model that does not require advanced technical infrastructure or real-time AI monitoring (Casal-Otero et al. 2023; Hough et al. 2025). This model is therefore feasible for implementation across diverse educational settings.

The choice of biomedical materials education as the context further strengthens the relevance of the findings. Sterilization methods constitute a domain where conceptual precision is paramount, and errors in understanding can have tangible clinical consequences. The success of the intervention in this context suggests that the AI-answer error-correction strategy may be particularly well suited to disciplines that require the integration of factual knowledge with applied reasoning (Guo et al. 2023; Metcalfe 2017). Nevertheless, the underlying pedagogical principle, which leverages students' critical evaluation of AI answers to promote critical evaluation, is broadly transferable (Dahri et al. 2024; Kasneci et al. 2023). Whether applied to pharmacology, biomedical engineering design, or clinical

decision making, the strategy offers a practical means of embedding critical thinking instruction within content-focused courses (Kasneci et al. 2023). Acknowledging the contextual boundaries of the present study, we explicitly address the generalizability of our findings beyond the specific biomedical materials course. While the intervention was implemented in a single disciplinary context with a relatively homogeneous student cohort, the core pedagogical principle, namely engaging learners in structured peer correction of AI-generated answers to promote critical evaluation and metacognitive regulation, is theoretically transportable to other educational settings. For instance, in disciplines such as pharmacology, clinical decision making, or engineering design, where conceptual precision and applied reasoning are equally paramount, the same strategy can be adapted by using domain-specific teacher-recommended and student-generated questions. Nevertheless, we caution that direct transfer requires appropriate contextual adaptation, including alignment with existing curricula, instructor training, and consideration of student AI literacy levels. Replication studies across diverse institutional contexts, learner populations, and subject areas are necessary to empirically verify the robustness and boundary conditions of the observed effects.

From a pedagogical standpoint, this study highlights the importance of intentional instructional design when incorporating generative AI into teaching. Simply granting students access to AI tools does not guarantee effective learning; indeed, unstructured use may reinforce uncritical acceptance (Kooli 2023; Ng et al. 2021). By contrast, the peer correction of AI answers to teacher-recommended and student-generated questions/statements provided explicit scaffolding for critical engagement, ensuring that students interacted with AI in a manner that fostered deeper cognitive processing. Additionally, the teacher summarization in the control group, while beneficial for knowledge consolidation, lacked the active error-driven cognitive conflict and question-generation component that appears central to metacognitive gains. This aligns with a growing consensus that AI literacy should be taught not as a standalone topic but as an integrated component of disciplinary learning, with structured activities that model thoughtful AI use (Ng et al. 2021).

5.1. Study limitations

Several limitations should be considered. First, the intervention was confined to a single 80-minute session addressing a discrete topic, limiting generalizability to broader curricular contexts. The durability of observed effects remains unknown, as no delayed post-test was administered to assess long-term retention or transfer to novel content. Metacognitive competencies typically require sustained reinforcement over time (Broadbent et al. 2015; Zimmerman 2002). Thus, longitudinal integration across multiple course units is necessary to determine whether benefits persist (Crompton et al. 2023; Gordon et al. 2024). Additionally, the brief intervention may not sufficiently capture the developmental trajectory of critical evaluation skills, which recent work suggests requires iterative exposure to AI-generated content across varied contexts (Casal-Otero et al. 2023).

Second, the quasi-experimental design with non-randomized assignment introduces potential selection bias, as students self-selected into groups based on schedule preferences. The lack of randomization limits causal inference, and the single-institution, single-cohort sample restricts generalizability to other learner populations or disciplines (Akgun et al. 2022; Zawacki-Richter et al. 2019). While extreme pre-test scores were excluded, unmeasured confounding variables such as prior motivation may have influenced outcomes. Replication across diverse institutional contexts and learner levels is needed to establish robustness (Ng et al. 2021; Sapci et al. 2020). Furthermore, self-report measures of metacognition may not fully capture actual regulatory behaviors during authentic tasks, and future studies should incorporate behavioral trace data or think-aloud protocols (Craig et al. 2020). Furthermore, we reflect critically on our statistical outcomes. Although the observed effect sizes, such as Cohen's $d = 0.910$ for AI learning attitudes, indicate practically meaningful differences, the relatively small sample size and the absence of a priori power analysis limit the precision of these estimates and elevate the risk of Type II error for non-significant trends. Readers are therefore advised to interpret the magnitude of effects with caution until replication studies with larger and adequately powered samples become available.

Finally, this study did not assess potential ethical risks or unintended consequences, such as over-reliance on AI tools or variations in AI literacy among learners (Hough et al. 2025; Kooli 2023; Liu et al. 2025a; Stahl et al. 2024). Moreover, the task of having students generate their own questions and critique AI answers is generally low risk, but careful instruction is needed to ensure that students do not simply accept AI answers without verification. The long-term effects of exposing students to AI-generated errors on trust calibration warrant further exploration. Future research employing longitudinal, multi-institutional designs with randomized allocation will strengthen causal evidence across the competency-based medical education continuum.

6. Conclusion

This quasi-experimental study provides preliminary evidence that a peer-driven, AI-answer error-correction strategy using teacher-recommended and student-generated questions/statements resulted in significantly higher scores in knowledge application and metacognitive awareness compared with a control condition combining AI self-learning with teacher summarization and key point reinforcement in biomedical materials education. By repositioning AI as a fallible answer generator for peer critique, the intervention engages learners in question formulation, error detection, and collaborative correction, processes that lie at the heart of self-regulated learning. These findings offer preliminary support for the idea that structured peer correction of AI answers could serve as a useful pedagogical strategy to foster critical thinking and metacognitive skills. Given the quasi-experimental design and the absence of delayed post-test, and the limitations discussed above (including small sample size, lack of randomization, and self-report measures), these conclusions should be interpreted cautiously. Further longitudinal research across multiple course units, with randomized allocation and objective behavioral measures, is warranted to assess the durability of these effects and their generalizability across diverse educational contexts.

List of abbreviations

AI, artificial intelligence; GenAI, generative artificial intelligence; LLMs, large language models; SD, standard deviations; CFA, confirmatory factor analysis; EFA, exploratory factor analysis; CFI, comparative fit index; RMSEA, root mean square error of approximation; SRMR, standardized root mean square residual; KMO, Kaiser-Meyer-Olkin.

CRedit authorship contribution statement

Siyyuan Yan: Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Yehao Yan:** Writing – original draft, Formal analysis, Data curation. **Shanshan Wang:** Investigation, Data curation. **Su Chen:** Formal analysis, Data curation. **Ying Li:** Investigation, Formal analysis. **Zhiying Ai:** Writing – review & editing, Visualization, Validation, Conceptualization.

Declaration of competing interest

The authors declare no competing interests

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijer.2026.103047](https://doi.org/10.1016/j.ijer.2026.103047).

References

- Akgun, S., & Greenhow, C. (2022). Artificial intelligence in education: Addressing ethical challenges in K-12 settings. *AI and Ethics*, 2(3), 431–440. <https://doi.org/10.1007/s43681-021-00096-7>
- Allison, J., Hwang, G. J., Mayer, R. E., et al. (2025). From generative AI to extended reality: Multidisciplinary perspectives on the challenges, opportunities, and future of educational computing. *Journal of Education and Computer Research*, 63(6), 1327–1363. <https://doi.org/10.1177/07356331251359964>
- Banihashem, S. K., Bond, M., Bergdahl, N., et al. (2025). A systematic mapping review at the intersection of artificial intelligence and self-regulated learning. *International Journal of Education and Technology H*, 22(1). <https://doi.org/10.1186/S41239-025-00548-8>
- Bond, M., Khosravi, H., De Laat, M., et al. (2024). A meta systematic review of artificial intelligence in higher education: a call for increased ethics, collaboration, and rigour. *International Journal of Education and Technology H*, 21(1). <https://doi.org/10.1186/s41239-023-00436-z>
- Broadbent, J., & Poon, W. L. (2015). Self-regulated learning strategies & academic achievement in online higher education learning environments: A systematic review. *The Internet and Higher Education*, 27, 1–13. <https://doi.org/10.1016/j.iheduc.2015.04.007>
- Casal-Otero, L., Catala, A., Fernandez-Morante, C., et al. (2023). AI literacy in K-12: a systematic literature review. *International Journal of STEM Education*, 10(1). <https://doi.org/10.1186/S40594-023-00418-7>
- Craig, K., Hale, D., Grainger, C., et al. (2020). Evaluating metacognitive self-reports: systematic reviews of the value of self-report in metacognitive research. *Metacognition and Learning*, 15(2), 155–213. <https://doi.org/10.1007/s11409-020-09222-y>
- Crompton, H., & Burke, D. (2023). Artificial intelligence in higher education: the state of the field. *International Journal of Education and Technology H*, 20(1). <https://doi.org/10.1186/S41239-023-00392-8>
- Dahri, N. A., Yahaya, N., Al-Rahmi, W. M., et al. (2024). Extended TAM based acceptance of AI-Powered ChatGPT for supporting metacognitive self-regulated learning in education: A mixed-methods study. *Heliyon*, 10(8), Article e29317. <https://doi.org/10.1016/j.heliyon.2024.e29317>
- Flavell, J. H. (1979). Metacognition and cognitive monitoring: A new area of cognitive-developmental inquiry. *American Psychologist*, 34(10), 906–911. <https://doi.org/10.1037/0003-066X.34.10.906>
- Gordon, M., Daniel, M., Ajiboye, A., et al. (2024). A scoping review of artificial intelligence in medical education: BEME Guide No. 84. *Medical Teacher*, 46(4), 446–470. <https://doi.org/10.1080/0142159X.2024.2314198>
- Guo, Y., & Lee, D. (2023). Leveraging ChatGPT for Enhancing Critical Thinking Skills. *Journal of Chemical Education*, 100(12), 4876–4883. <https://doi.org/10.1021/acs.jchemed.3c00505>
- Hough, J., Culley, N., Erganian, C., et al. (2025). Potential risks of GenAI on medical education. *Bmj Evid-Based Medicine*, 30(6), 406–408. <https://doi.org/10.1136/bmjebm-2025-114339>
- Joo, Y. J., So, H. J., & Kim, N. H. (2018). Examination of relationships among students' self-determination, technology acceptance, satisfaction, and continuance intention to use K-MOOCs. *Computers & Education*, 122, 260–272. <https://doi.org/10.1016/j.compedu.2018.01.003>
- Kapur, M. (2008). Productive failure. *Cognition Instruct*, 26(3), 379–424. <https://doi.org/10.1080/07370000802212669>
- Karaoglan Yilmaz, F. G., Yilmaz, R., Ustun, A. B., et al. (2025). Exploring the role of cognitive flexibility, digital competencies, and self-regulation skills on students' generative artificial intelligence anxiety. *Computers in Human Behavior: Artificial Humans*, 5, Article 100187. <https://doi.org/10.1016/j.chbah.2025.100187>
- Kasneci, E., Sessler, K., Küchemann, S., et al. (2023). ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and Individual Differences*, 103, Article 102274. <https://doi.org/10.1016/j.lindif.2023.102274>
- Kim, J., & Lee, S. S. (2023). Are Two Heads Better Than One?: The Effect of Student-AI Collaboration on Students' Learning Task Performance. *TechTrends: for Leaders in Education & Training*, 67(2), 365–375. <https://doi.org/10.1007/s11528-022-00788-9>
- Kooli, C. (2023). Chatbots in Education and Research: A Critical Examination of Ethical Implications and Solutions. *Sustainability-Basel*, 15(7), 5614. <https://doi.org/10.3390/Su15075614>
- Liu, M., Cheng, Y., Li, S., et al. (2025a). Exploring medical students' attitudes and perceptions toward artificial intelligence in medicine in Shandong Province, China. *BMC Medical Education*, 26(1), 106. <https://doi.org/10.1186/s12909-025-08465-0>
- Liu, X. H., Xiao, Y. Y., & Li, D. L. (2025b). Assessing strategic use of artificial intelligence in self-regulated learning: Instrument development and evidence from Chinese university students. *International Journal of Education and Technology H*, 22(1). <https://doi.org/10.1186/S41239-025-00567-5>
- Luo, J. H., Zheng, C. X., Yin, J. M., et al. (2025). Design and assessment of AI-based learning tools in higher education: a systematic review. *International Journal of Education and Technology H*, 22(1). <https://doi.org/10.1186/S41239-025-00540-2>
- Martín-Moncunill, D., & Alonso Martínez, D. (2025). Students' Trust in AI and Their Verification Strategies: A Case Study at Camilo José Cela University. *Education Sciences*, 15(10), 1307. <https://doi.org/10.3390/educsci15101307>
- McPhee, S. W., & Jerowsky, M. (2025). Beyond technical skills: a pedagogical perspective on fostering critical engagement with generative AI in university classrooms. *Frontiers in Education*, 10. <https://doi.org/10.3389/feduc.2025.1593278>
- Metcalfe, J. (2017). Learning from Errors. *Annual Review of Psychology*, 68, 465–489. <https://doi.org/10.1146/annurev-psych-010416-044022>

- Moreno, E., Alvarez-Lozada, L. A., Arrambide-Garza, F. J., et al. (2023). ChatGPT utility in healthcare education, research, and practice: systematic review on the promising perspectives and valid concerns. *Healthcare*, 11(21), 2819. <https://doi.org/10.3390/Healthcare11212819>
- Ng, D. T. K., Leung, J. K. L., Chu, S. K. W., et al. (2021). Conceptualizing AI literacy: An exploratory review. *Computers and Education: Artificial Intelligence*, 2(c), Article 100041. <https://doi.org/10.1016/j.caeai.2021.100041>
- Nussbaum, M., Bekerman, Z., Archambault, L., et al. (2025). In search of adequate models of pedagogy. *Computers & Education*, 235, Article 105345. <https://doi.org/10.1016/j.compedu.2025.105345>
- Nussbaum, M., Bekerman, Z., Archambault, L., et al. (2026). Teachers' doings in the AI era: From transmission to relation. *Computers & Education*, 249, Article 105617. <https://doi.org/10.1016/j.compedu.2026.105617>
- Sapci, A. H., & Sapci, H. A. (2020). Artificial Intelligence Education and Tools for Medical and Health Informatics Students: Systematic Review. *JMIR Medical Education*, 6(1), Article e19285e19285. <https://doi.org/10.2196/19285>
- Schraw, G., & Dennison, R. S. (1994). Assessing Metacognitive Awareness. *Contemporary Educational Psychology*, 19(4), 460–475. <https://doi.org/10.1006/ceps.1994.1033>
- Simsek, A. S., Cengiz, G. S. T., & Bal, M. (2025). Extending the TAM framework: Exploring learning motivation and agility in educational adoption of generative AI. *Education of Information and Technology*, 30(15), 20913–20942. <https://doi.org/10.1007/s10639-025-13591-9>
- Sinha, T., & Kapur, M. (2021). When Problem Solving Followed by Instruction Works: Evidence for Productive Failure. *Review of Educational Research*, 91(5), 761–798. <https://doi.org/10.3102/00346543211019105>
- Stahl, B. C., & Eke, D. (2024). The ethics of ChatGPT - Exploring the ethical issues of an emerging technology. *International Journal of Information Management*, 74, Article 102700. <https://doi.org/10.1016/j.ijinfomgt.2023.102700>
- Tu, Y.-F., & Lu, Y.-C. (2025). Trends of generative AI applications in educational settings. *International Journal of Mobile Learning and Organisation*, 19(4), 442–467. <https://doi.org/10.1504/ijmlo.2025.149024>
- Wong, S. S. H., & Lim, S. W. H. (2022). Deliberate Errors Promote Meaningful Learning. *Journal of Educational Psychology*, 114(8), 1817–1831. <https://doi.org/10.1037/edu0000720>
- Yilmaz, F. G. K., & Yilmaz, R. (2025). Exploring the role of self-regulated learnings skills, cognitive flexibility, and metacognitive awareness on generative artificial intelligence attitude. *Innov Education Teaching International*, 62(5), 1682–1695. <https://doi.org/10.1080/14703297.2025.2484613>
- Zawacki-Richter, O., Marin, V. I., Bond, M., et al. (2019). Systematic review of research on artificial intelligence applications in higher education - where are the educators? *International Journal of Education and Technology H*, 16(1). <https://doi.org/10.1186/s41239-019-0171-0>
- Zimmerman, B. J. (2002). Becoming a self-regulated learner: An overview. *Theor Pract*, 41(2), 64–70. https://doi.org/10.1207/s15430421tip4102_2